Database Group Project Requirements

Key Focus Areas:

- Database Design & Normalization (3NF)
- Performance Analysis & Optimization
- Data Integrity & Security
- Advanced SQL Operations
- Audit Mechanisms & Triggers
- Transaction Management

Project Deliverables

- Functional Requirement Document (FRD)
- Technical Design Document (TDD)
- Data Model / ER Diagram
- Schema Build File
- Script: Extract from repo & build schema with initial data

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Architecture & Design Specifications:

- **System Architecture:** Database tier, application layer integration
- **Technology Stack:** DBMS selection justification
- **Data Flow Diagrams:** Information movement through system
- **Security Architecture:** Access control, encryption, audit logging
- **Performance Considerations:** Indexing strategy, query optimization
- **Scalability Planning:** Growth accommodation strategies

Entity-Relationship Design Requirements:

- **Minimum 8-10 entities** with meaningful relationships
- **All relationship types:** One-to-One, One-to-Many, Many-to-Many
- **Complete attribute specification:** Primary keys, foreign keys, data types
- **Cardinality notation:** Clearly marked relationship cardinalities
- **Weak entities:** At least one weak entity with identifying relationship

ER Diagram Components:

- Entities (rectangles) with all attributes listed
- Relationships (diamonds) with descriptive names
- Cardinality constraints (1:1, 1:N, M:N)
- Participation constraints (total/partial)
- Attribute types (key, derived, multi-valued, composite)

Mapping Documentation:

ER Component	Database Implementation	Constraints Applied
Strong Entity	Table with Primary Key	NOT NULL, UNIQUE constraints
Weak Entity	Table with Composite Key	Foreign Key + Partial Key
1:N Relationship	Foreign Key in "many" side	Referential Integrity
M:N Relationship	Junction/Bridge Table	Composite Primary Key

Triggers & Constraints Implementation:

- **CHECK constraints:** Data validation rules
- **FOREIGN KEY constraints:** Referential integrity
- **UNIQUE constraints:** Data uniqueness
- **BEFORE/AFTER triggers:** Business rule enforcement

Demonstrations

- Transaction killed and rolled back
- Audit mechanism capturing changes
- Trigger making an update
- Constraint enforcement

SQL Queries to Demonstrate

- Joins
- Updates
- Subqueries
- Aggregates
- Exception handling

Normalization, Entitlements & Performance

- Database demonstrated to be in 3NF
- Security model with role definitions and entitlements based on roles

Performance considerations:

- Large batch transactions
- Reads
- Updates and Deletes
- Large volume of transactions
- Large data extracts